In [1]:
```python
#tuple
t=()
t
```
Out[1]:
```
()
```
In [2]:
```python
type(t)
```
Out[2]:
```
tuple
```
In [8]:
```python
t1=(10,20.30)
t1
```
Out[8]:
```
(10, 20.3)
```
In [5]:
```python
t2=(10,2.3,'nit',True,1+2j)
t2
```
Out[5]:
```
(10, 2.3, 'nit', True, (1+2j))
```
In [6]:
```python
t1.count(10)
```
Out[6]:
```
1
```
In [10]:
```python
t1.index(10)
```
Out[10]:
```
0
```
In [11]:
```python
t1.append(10)
```
```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[11], line 1
----> 1 t1.append(10)

AttributeError: 'tuple' object has no attribute 'append'
```
In [12]:
```python
t1[0]
```
Out[12]:
```
10
```
In [13]:
```python
t1[0]=100
```
```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[13], line 1
----> 1 t1[0]=100

TypeError: 'tuple' object does not support item assignment
```
In [15]:
```python
icici=('john',123456,'cizps')
icici
```
Out[15]:
```
('john', 123456, 'cizps')
```
In [16]:
```python
icici[1]=3456
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[16], line 1
----> 1 icici[1]=3456

TypeError: 'tuple' object does not support item assignment
```

In [17]:
```python
t1.remove(20)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[17], line 1
----> 1 t1.remove(20)

AttributeError: 'tuple' object has no attribute 'remove'
```

In [19]:
```python
t3=t1*3
t3
```

Out[19]:
```
(10, 20.3, 10, 20.3, 10, 20.3)
```

In [22]:
```python
#set
s={}
s
```

Out[22]:
```
{}
```

In [23]:
```python
type(s)
```

Out[23]:
```
dict
```

In [29]:
```python
s1 = {100, 4, 10, 3, 90, 20, 50}
s1
print(type(s1))
```

```
<class 'set'>
```

In [31]:
```python
s2={3,1+2j,True,3.14}
s2
```

Out[31]:
```
{(1+2j), 3, 3.14, True}
```

In [32]:
```python
s1.add(25)
s1
```

Out[32]:
```
{3, 4, 10, 20, 25, 50, 90, 100}
```

In [33]:
```python
s1.add(200,300)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[33], line 1
----> 1 s1.add(200,300)

TypeError: set.add() takes exactly one argument (2 given)
```

In [35]:
```python
s3=s1.copy()
s3
```

Out[35]: {3, 4, 10, 20, 25, 50, 90, 100}

In [37]:
```
s3.clear()
s3
```

Out[37]: set()

In [38]:
```
s1[0]
s1
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[38], line 1
----> 1 s1[0]
      2 s1

TypeError: 'set' object is not subscriptable
```

In [39]:
```
s1
```

Out[39]: {3, 4, 10, 20, 25, 50, 90, 100}

In [40]:
```
s1.pop()
```

Out[40]: 50

In [41]:
```
s1.pop()
```

Out[41]: 3

In [42]:
```
s1.pop(2)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[42], line 1
----> 1 s1.pop(2)

TypeError: set.pop() takes no arguments (1 given)
```

In [45]:
```
s1.discard(25)
s1
```

Out[45]: {4, 10, 20, 90, 100}

In [46]:
```
s1.remove()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[46], line 1
----> 1 s1.remove()

TypeError: set.remove() takes exactly one argument (0 given)
```

In [47]:
```

```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[47], line 1
----> 1 tup1()

NameError: name 'tup1' is not defined
```

In [48]:
```
#nested tuple
tup5=('honey',25,(50,100),(200,300))
tup5
```

Out[48]: ('honey', 25, (50, 100), (200, 300))

In [65]:
```
#tuple of mixed data type
tup4=(100,'asif',18.90)
tup4
```

Out[65]: (100, 'asif', 18.9)

In [51]:
```
len(tup4)
```

Out[51]: 3

In [72]:
```
#tuple slicing
tup5=('one','two','three','four','five','six','seven','eight')
```

In [53]:
```
tup5[0:3]
```

Out[53]: ('one', 'two', 'three')

In [55]:
```
tup5[:3]
```

Out[55]: ('one', 'two', 'three')

In [57]:
```
tup5[-3:]
```

Out[57]: ('six', 'seven', 'eight')

In [59]:
```
tup5[-1]
```

Out[59]: 'eight'

In [60]:
```
del tup5
```

In [61]:
```
tup5
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[61], line 1
----> 1 tup5

NameError: name 'tup5' is not defined
```

In [62]:
```
 del tup4 [0]
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[62], line 1
----> 1 del tup4 [0]

TypeError: 'tuple' object doesn't support item deletion
```

In [63]:
```python
tup4[4]=1
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[63], line 1
----> 1 tup4[4]=1

TypeError: 'tuple' object does not support item assignment
```

In [66]:
```python
#loop
for i in tup4:
    print(i)
```

```
100
asif
18.9
```

In [70]:
```python
for i in enumerate(tup4):
print(i)
```

```
  Cell In[70], line 2
    print(i)
    ^
IndentationError: expected an indented block after 'for' statement on line 1
```

In [73]:
```python
#tuple membersip
tup5
```

Out[73]: `('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')`

In [74]:
```python
'one' in tup5
```

Out[74]: `True`

In [76]:
```python
if 'three' in tup5:
    print('preseent')
else:
    print('not present')
```

```
preseent
```

In [78]:
```python
#index positions
tup5.index('one')
```

Out[78]: `0`

In [79]:
```python
#sorting
tup6=(43,67,99,12,8,90,67)
tup6
```

Out[79]: `(43, 67, 99, 12, 8, 90, 67)`

In [80]:
```
sorted(tup6)
```
Out[80]:
```
[8, 12, 43, 67, 67, 90, 99]
```
In [81]:
```
tup6.sort()
```
```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[81], line 1
----> 1 tup6.sort()

AttributeError: 'tuple' object has no attribute 'sort'
```
In [82]:
```
sorted(tup6,reverse=True)
```
Out[82]:
```
[99, 90, 67, 67, 43, 12, 8]
```